Exercise 9. Show that the determinant of the $n \times n$ *companion matrix*

$$C_a = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ \vdots & & 1 & & \vdots \\ & & \ddots & \ddots & 0 \\ 0 & & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix},$$

where $a_0, a_1, \ldots, a_{n-1} \in \mathbb{C}$, is equal to $(-1)^n a_0$.

Hint. Multiply the ith row ($i = 1, 2, \ldots, n-1$) by a_i and add it to the last one.

Exercise 10. If $A = [a_{ij}]_{i,j=1}^{m,n}$, $m \leq n$, and $B = [b_{ij}]_{i,j=1}^{n,m}$, check that

$$\det \begin{bmatrix} A & 0 \\ -I_n & B \end{bmatrix} = \det \begin{bmatrix} A & AB \\ -I_n & 0 \end{bmatrix},$$

where I_n stands for the $n \times n$ identity matrix.

Hint. Use Property 5 and add b_{jk} times the jth column ($1 \leq j \leq n$) of the matrix on the left to the $(n+k)$th column, $k = 1, 2, \ldots, m$. □

2.3 Cofactor Expansions

Another method for computation of determinants is based on their reduction to determinants of matrices of smaller sizes. This is in contrast to the method of Gauss (see Example 2.2.4). The following notion is important for this approach.

Let A be an $n \times n$ matrix. A *minor* of order $n-1$ of A is defined to be the determinant of a submatrix of A obtained by striking out one row and one column from A. The minor obtained by striking out the ith row and jth column is written M_{ij} $(1 \leq i, j \leq n)$.

Exercise 1. Concerning the companion matrix of order 4 in Exercise 2.2.9,

$$M_{11} = \det \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} = -a_1,$$

$$M_{41} = \det \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1. \quad \square$$

鉴于如下结论，子式在行列式的计算中可发挥重要作用。

定理 1（代数余子式展开式，Cofactor expansion）．设 A 为任意一个 n × n 矩阵。则对任意 i , j (1 ≤ i , j ≤ n)，

$$C_a = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ \vdots & & 1 & & \vdots \\ & & \ddots & \ddots & 0 \\ 0 & & \cdots & 0 & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix},$$

o r ,；类似地，

$$\det\begin{bmatrix} A & 0 \\ -I_n & B \end{bmatrix} = \det\begin{bmatrix} A & AB \\ -I_n & 0 \end{bmatrix},$$

其中 A _(pq) = (- 1) ^(p+q) M _(pq)。

在深入展开证明的细节之前，请注意：数值 A _(pq) (1 ≤ p , q ≤ n) 被称为元素 a _(pq) 的代数余子式，因此公式（1）与（2）分别被称为行列式 det A 按第 i 行与按第 j 列的代数余子式展开式。于是，代数余子式 A _(pq) 恰为子式 M _(pq) 乘以 +1 或 −1，其符号由下标之和 i + j 的奇偶性决定：若 i + j 为偶数，则取正号；若为奇数，则取负号。

证明．证明始于如下观察：det A 的表达式（2.1.8）中的每一项均包含第 i 行的一个元素。因此，将所有含元素 a _(ij) (j = 1 , 2 , … , n) 的项归并起来，我们得到

$$M_{11} = \det\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_1 & -a_2 & -a_3 \end{bmatrix} = -a_1,$$

故为证公式（1），只需证明对所有 i , j .，均有

$$M_{41} = \det\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1. \quad \square$$

首先考虑情形 i = j = 1；我们来证明 A □□ = A □□。事实上，由（2.1.8）式及（3）式中对 A □□ 的构造可知，

其中 t (j) 表示排列 (1 , j □ , j □ , ... j _n) 的逆序数，等价地，也表示排列 (j □ , j □ , … , j _n) 关于数集 2 , 3 , … , n . 的逆序数。

利用行列式定义，对矩阵 A 删除第一行与第一列后所得的 (n - 1) × (n - 1)
阶矩阵计算其行列式，可立即得出：式 (5) 中的表达式即为此行列式，亦即 M □□。因此，A □□ = M □□ = (- 1) ¹□¹ M = A □□。
为在一般情形下证明式 (4)，我们首先通过 i - 1 次相邻行的两两对换，再继之以 j - 1
次相邻列的两两对换，将元素 a _(ij) 移至位置 (1,1)。记重排后的矩阵为 B。由于对应于余子式 a _(ij)
的子矩阵的各行、各列在 A 与 B
中的相对位置均未改变（即该子矩阵本身所含行指标与列指标的相对顺序保持不变），故 a _(ij) 在 A 与 B
中所对应的余子式相同。因此，应用前述已证的特殊情形，有

Minors can play an important role in computing the determinant, in view of the following result.

Theorem 1 (Cofactor expansion). *Let A be an arbitrary $n \times n$ matrix. Then for any i, j $(1 \leq i, j \leq n)$;*

$$\det A = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \tag{1}$$

or, similarly,

$$\det A = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}, \tag{2}$$

where $A_{pq} = (-1)^{p+q}M_{pq}$.

Before proceeding with the details of the proof, note that the numbers A_{pq} $(1 \leq p, q \leq n)$ are called the *cofactors* of the elements a_{pq} and, therefore, formulas (1) and (2) are referred to as, respectively, *row* and *column cofactor expansions* of $\det A$. Thus, the cofactor A_{pq} is just the minor M_{pq} multiplied by $+1$ or -1, according to whether the sum of the subscripts is even or odd.

PROOF. The proof begins with the observation that every term in Eq. (2.1.8) for $\det A$ contains an element of the ith row. Therefore, collecting together the terms containing a_{ij} $(j = 1, 2, \ldots, n)$, we obtain

$$\det A = a_{i1}\tilde{A}_{i1} + a_{i2}\tilde{A}_{i2} + \cdots + a_{in}\tilde{A}_{in} \qquad (1 \leq i \leq n). \tag{3}$$

Hence, to establish (1), it suffices to show that

$$\tilde{A}_{ij} = A_{ij} \tag{4}$$

for all i, j.

Consider first the case $i = j = 1$; we show that $\tilde{A}_{11} = A_{11}$. Indeed, from (2.1.8) and the construction of $\tilde{A}_{11}$ in (3), it follows that

$$\tilde{A}_{11} = \sum_{j} (-1)^{t(j)} a_{2j_2} a_{3j_3} \cdots a_{nj_n}, \tag{5}$$

where $t(j)$ denotes the number of inversions of the permutation $(1, j_2, j_3, \ldots j_n)$ or, equivalently, of permutation $(j_2, j_3, \ldots, j_n)$ of the numbers $2, 3, \ldots, n$.

Using the definition of the determinant for the $(n-1) \times (n-1)$ matrix obtained by deleting the first row and the first column of A, it is easily concluded that the expression in (5) is that determinant, that is, M_{11}. Thus $\tilde{A}_{11} = M_{11} = (-1)^{1+1}M_{11} = A_{11}$.

To prove (4) in the general case, we first shift a_{ij} to the (1, 1) position by means of $i - 1$ successive interchanges of adjacent rows followed by $j - 1$ interchanges of adjacent columns. Call the rearranged matrix B. The minor associated with a_{ij} is the same in both A and B because the relative positions

$$\det A = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}$$

但行列式的性质 3 表明：

$$\det A = a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj},$$

故有：

$$\det A = a_{i1}\tilde{A}_{i1} + a_{i2}\tilde{A}_{i2} + \cdots + a_{in}\tilde{A}_{in} \qquad (1 \leq i \leq n).$$

将上式与式 (3) 比较，即得所求结论。式 (2) 则可由式 (1) 及等式 □ _A = det □ _A ^T 直接推出。

习题 2．矩阵的行列式

M _(ij)

$$\tilde{A}_{ij} = A_{ij}$$

可借助沿第三行的代数余子式展开式简便地计算。事实上，第三行仅含一个非零元素，因此根据（1）式，有：

$$\tilde{A}_{11} = \sum_{j} (-1)^{t(j)} a_{2j_2} a_{3j_3} \cdots a_{nj_n},$$

下一习题表明：若在表达式（1）中，将代数余子式 A _ { i j } (1 \leq i \leq n , j = 1 , 2 , . . . , n)
替换为对应于另一行元素的代数余子式，则所得表达式恒等于零。对列亦有类似结论。

习题 3．证明：若 i \neq r ,，则

且若 \operatorname { i f } j \neq s ,，则

提示：考虑将矩阵 A 的第 i 行替换为第 r 行 (i \neq r) 后所得的矩阵，并利用性质 4。

of rows and columns of the submatrix corresponding to this minor M_{ij} are unchanged. Hence, using the special case already proved,

$$\det B = a_{ij}M_{ij} + (\text{terms not involving } a_{ij}).$$

But Property 3 of determinants implies that

$$\det B = (-1)^{i-1}(-1)^{j-1} \det A = (-1)^{i+j} \det A.$$

Hence

$$\det A = (-1)^{i+j}a_{ij}M_{ij} + (\text{terms not involving } a_{ij}).$$

The result now follows on comparison with (3). Equation (2) can now be obtained using Eq. (1) and the fact that $\det A = \det A^{\mathrm{T}}$. ■

Exercise 2. The determinant of the matrix

$$A = \begin{bmatrix} 3 & 12 & -1 & 0 \\ -1 & 0 & 5 & 6 \\ 0 & 3 & 0 & 0 \\ 0 & 4 & -1 & 2 \end{bmatrix}$$

can be easily evaluated by use of the cofactor expansion along the third row. Indeed, the third row contains only one nonzero element and, therefore, in view of (1);

$$\det A = 3A_{32} = 3(-1)^{3+2} \det \begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & 6 \\ 0 & -1 & 2 \end{bmatrix} = (-3)46 = -138. \quad \square$$

The next exercise shows that if in expression (1) the cofactors A_{ij} $(1 \leq i \leq n, j = 1, 2, \ldots, n)$ are replaced by those corresponding to elements of another row, then the expression obtained is equal to zero. A similar result holds for columns.

Exercise 3. Show that if $i \neq r$, then

$$a_{i1}A_{r1} + a_{i2}A_{r2} + \cdots + a_{in}A_{rn} = 0, \tag{6}$$

and if $j \neq s$, then

$$a_{1j}A_{1s} + a_{2j}A_{2s} + \cdots + a_{nj}A_{ns} = 0. \tag{7}$$

Hint. Consider the matrix obtained from A by replacing row i by row r $(i \neq r)$ and use Property 4.

习题 4．验证：A ,，其中 A 是 n \times n 阶斐波那契矩阵

$$\det B = a_{ij}M_{ij} + (\text{terms not involving } a_{ij}).$$

恰好是斐波那契数列的第 n 项。

$$\det B = (-1)^{i-1}(-1)^{j-1} \det A = (-1)^{i+j} \det A.$$

其中 a _n = a _(n-1) + a _(n-2) (n ≥ 3) 。

习题 5.（斐波那契矩阵是一类“三对角”矩阵的实例。本习题讨论一般的三对角矩阵。）令 J _n 表示 n 阶雅可比矩阵；

$$\det A = (-1)^{i+j}a_{ij}M_{ij} + (\text{terms not involving } a_{ij}).$$

证明：

$$A = \begin{bmatrix} 3 & 12 & -1 & 0 \\ -1 & 0 & 5 & 6 \\ 0 & 3 & 0 & 0 \\ 0 & 4 & -1 & 2 \end{bmatrix}$$

其中缩写符号 | A |，以及 A，均表示矩阵 A 的行列式。

习题 6. 验证：

$$\det A = 3A_{32} = 3(-1)^{3+2} \det \begin{bmatrix} 3 & -1 & 0 \\ -1 & 5 & 6 \\ 0 & -1 & 2 \end{bmatrix} = (-3)46 = -138.$$

式 (8) 中的矩阵，记作 V _n , ，称为 n 阶范德蒙德矩阵。式 (8)
所给出的关系表明：该矩阵的行列式等于所有形如 x _i - x _j (1 ≤ j < i ≤ n ）的差之乘积。

提示：利用第 2.2 节的性质 5，并将第 i 行乘以 - ±b x □ 后加到第 i + 1 (i = n - 1 , n - 2 , … , 1 ）行上。再应用列方向的代数余子式展开公式，将 det V _n 约化为 det V _(n-1)。

Exercise 4. Check that det A, where A is the $n \times n$ *Fibonacci matrix*

$$A = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ -1 & 1 & 1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix},$$

is exactly the nth term in the Fibonacci sequence

$$1, 2, 3, 5, 8, 13, \ldots = \{a_n\}_{n=1}^{\infty},$$

where $a_n = a_{n-1} + a_{n-2}$ $(n \geq 3)$.

Exercise 5. (The Fibonacci matrix is an example of a "tridiagonal" matrix. This exercise concerns general tridiagonal matrices.) Let J_n denote the *Jacobi matrix* of order n;

$$J_n = \begin{bmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_1 & a_2 & b_2 & & \vdots \\ 0 & c_2 & a_3 & b_3 & 0 \\ \vdots & & \ddots & \ddots & b_{n-1} \\ 0 & & \cdots & c_{n-1} & a_n \end{bmatrix}.$$

Show that

$$|J_n| = a_n|J_{n-1}| - b_{n-1}c_{n-1}|J_{n-2}| \qquad (n \geq 3),$$

where the abbreviation $|A|$, as well as det A, denotes the determinant of A.

Exercise 6. Confirm that

$$\det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & & x_n \\ x_1^2 & x_2^2 & & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{bmatrix} = \prod_{1 \leq j < i \leq n} (x_i - x_j). \tag{8}$$

The matrix in (8), written V_n, is called the *Vandermonde matrix* of order n. The relation in (8) states that its determinant is equal to the product of all differences of the form $x_i - x_j$ $(1 \leq j < i \leq n)$.

Hint. Use Property 5 of Section 2.2 and add the ith row multiplied by $-x_1$ to the row $i + 1$ $(i = n - 1, n - 2, \ldots, 1)$. Apply a column cofactor expansion formula for the reduction of det V_n to det V_{n-1}.

习题 7．设 C _a 表示与（首一）多项式

$$A=\begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ -1 & 1 & 1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix},$$

相关联的友矩阵（参见习题 2.2.9），证明

$$1, 2, 3, 5, 8, 13, \ldots = \{a_n\}_{n=1}^{\infty},$$

由此推出：det (λ □ I - C _a) = 0 当且仅当 λ □ 是 ±b a (λ) 的一个零点。□

2.4 拉普拉斯定理

行列式的行方向与列方向代数余子式展开可借助 p 阶子式的思想自然地推广。回顾一下，我们此前已定义了 n × n 行列式的 p 阶子式。† 更一般地，若 A 是一个 m × n 矩阵，则从 A 中删去 m-p 行与 n-p 列后所得的 p × p 子矩阵的行列式，称为 A 的一个 p 阶子式。

更具体地说，若所保留的行与列的下标分别为

$$J_n=\begin{bmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_1 & a_2 & b_2 & & \vdots \\ 0 & c_2 & a_3 & b_3 & 0 \\ \vdots & & \ddots & \ddots & b_{n-1} \\ 0 & & \cdots & c_{n-1} & a_n \end{bmatrix}.$$

则相应的 p 阶子式记为 p × p。

p (1 ≤ p ≤ n)

n - 1

A (1 ≤ p ≤ min (m , n))

m - p

n - p

$$|J_n| = a_n|J_{n-1}| - b_{n-1}c_{n-1}|J_{n-2}| \qquad (n \geq 3),$$

例如，一个 n \times n 阶矩阵 A 的阶为 n - 1 的子式 M _ { 1 n }，在此记号下为

$$\det\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & & x_n \\ x_1^2 & x_2^2 & & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{bmatrix} = \prod_{1 \leq j < i \leq n} (x_i - x_j).$$

而一阶子式即为该矩阵的各个元素。

满足 i _ { k } = j _ { k } (k = 1 , 2 , . . . , p) 的子式称为矩阵 A 的 p , 阶主子式；而满足 i _ { k } = j _ { k } = k (k = 1 , 2 , . . . , p) 的子式则称为矩阵 A 的首主子式（或顺序主子式）。

在将公式 (2.3.1) 与 (2.3.2) 推广至一般情形之前，我们还需引入余子式（complementary minor）的概念：它是指从方阵 A 中删去 (1) 式所列出行号与列号后所得子矩阵的行列式。我们将其记作

于是，对应于 (2) 式的余代数余子式（complementary cofactor）定义为

***Exercise* 7.** Show that if C_a denotes the companion matrix (see Exercise 2.2.9) associated with the (monic) polynomial

$$a(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1} + \lambda^n,$$

then

$$\det(\lambda I - C_a) = a(\lambda).$$

Deduce that $\det(\lambda_0 I - C_a) = 0$ if and only if λ_0 is a zero of $a(\lambda)$. □

2.4 Laplace's Theorem

The row and column cofactor expansions of a determinant can be generalized in a natural way, using the idea of a minor of order p $(1 \le p \le n)$. Recall that we have already defined minors of order $n - 1$ of an $n \times n$ determinant.† More generally, if A is an $m \times n$ matrix, then the determinant of a $p \times p$ submatrix of A $(1 \le p \le \min(m, n))$, obtained from A by striking out $m - p$ rows and $n - p$ columns, is called a *minor of order p* of A.

In more detail, if the rows and columns *retained* are given by subscripts

$$1 \le i_1 < i_2 < \cdots < i_p \le m, \qquad 1 \le j_1 < j_2 < \cdots < j_p \le n, \tag{1}$$

respectively, then the corresponding $p \times p$ minor is denoted by

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} \triangleq \det[a_{i_k j_k}]_{k=1}^{p}. \tag{2}$$

For example, the minor M_{1n} of order $n - 1$ of an $n \times n$ matrix A is, in this notation,

$$M_{1n} = A\begin{pmatrix} 2 & 3 & \cdots & n \\ 1 & 2 & \cdots & n-1 \end{pmatrix},$$

and the minors of order one are merely the elements of the matrix.

The minors for which $i_k = j_k$ $(k = 1, 2, \ldots, p)$ are called the *principal minors* of A of order p, and the minors with $i_k = j_k = k$ $(k = 1, 2, \ldots, p)$ are referred to as the *leading principal minors* of A.

Before proceeding to a generalization of Eqs. (2.3.1) and (2.3.2), we also need the notion of *complementary minors*, that is, the determinant of the sub-

† That is, the determinant of an $n \times n$ matrix.

其中，s = (i □ + i □ + ·s + i _p) + (j □ + j □ + ·s + j _p) .。显然，当 p = 1 时，这一概念即退化为此前所定义的矩阵元素的代数余子式。需注意：仅由指标 ^(is) c ^s 的位置即可判别我们所考虑的是余子式（位于右侧）还是代数余子式（位于左侧）。

例 1．考虑一个 5 × 5 阶矩阵 A = [a _(ij)] _(i,j=1) ⁵。则例如有：

$$a(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1} + \lambda^n,$$

$$\det(\lambda I - C_a) = a(\lambda).$$

$$1 \le i_1 < i_2 < \cdots < i_p \le m, \qquad 1 \le j_1 < j_2 < \cdots < j_p \le n,$$

其中，s = (2 + 3 + 5) + (1 + 2 + 4) = 1 7 .。

定理 1（拉普拉斯定理）．设 A 为任意一个 n × n 阶矩阵，并任取 A 的 p 行（或 p 列）。则 A 的行列式 det A 等于所有位于这些行（或列）上的 C _(n,p) 阶子式† 与其相应余子式的乘积之和。

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} \triangleq \det[a_{i_k j_k}]_{k=1}^{p}.$$

其中求和遍历所有 C _ { n , p } 个互异的（列）指标集 j _ { 1 } , j _ { 2 } , \ldots , j _ { p } \left(1 \leq j _ { 1 } < j _ { 2 } < \cdots < j _ { p } \leq n \right)。等价地，若改用列来表述，则有

$$M_{1n} = A\begin{pmatrix} 2 & 3 & \cdots & n \\ 1 & 2 & \cdots & n-1 \end{pmatrix},$$

其中 1 \leq i _ { 1 } < i _ { 2 } < \cdots < i _ { p } \leq n .。

matrix of a *square* matrix A resulting from the *deletion* of the rows and columns listed in (1). We denote it by

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}^{\mathrm{c}}.$$

The *complementary cofactor* to (2) is then defined by

$$A^{\mathrm{c}}\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} \triangleq (-1)^s A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}^{\mathrm{c}},$$

where $s = (i_1 + i_2 + \cdots + i_p) + (j_1 + j_2 + \cdots + j_p)$. Obviously, in the case $p = 1$, this notion coincides with the previously defined cofactor of an element. Note that only the position of the index "c" indicates whether we are considering a complementary minor (on the right) or complementary cofactor (on the left).

Example 1. Consider a 5×5 matrix $A = [a_{ij}]_{i,j=1}^{5}$. Then, for instance,

$$A\begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \end{pmatrix} = \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{51} & a_{52} & a_{54} \end{vmatrix},$$

$$A\begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \end{pmatrix}^{\mathrm{c}} = A\begin{pmatrix} 1 & 4 \\ 3 & 5 \end{pmatrix} = \begin{vmatrix} a_{13} & a_{15} \\ a_{43} & a_{45} \end{vmatrix},$$

$$A^{\mathrm{c}}\begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \end{pmatrix} = (-1)^s \begin{vmatrix} a_{13} & a_{15} \\ a_{43} & a_{45} \end{vmatrix} = -\begin{vmatrix} a_{13} & a_{15} \\ a_{43} & a_{45} \end{vmatrix},$$

where $s = (2 + 3 + 5) + (1 + 2 + 4) = 17$. □

Theorem 1 (Laplace's theorem). *Let A denote an arbitrary $n \times n$ matrix and let any p rows (or columns) of A be chosen. Then* det A *is equal to the sum of the products of all $C_{n,p}$ minors† lying in these rows with their corresponding complementary cofactors*:

$$\det A = \sum_j A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} A^{\mathrm{c}}\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}, \tag{3}$$

where the summation extends over all $C_{n,p}$ distinct sets of (column) indices $j_1, j_2, \ldots, j_p$ $(1 \le j_1 < j_2 < \cdots < j_p \le n)$. Or, equivalently, using columns,

$$\det A = \sum_i A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} A^{\mathrm{c}}\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}. \tag{4}$$

where $1 \le i_1 < i_2 < \cdots < i_p \le n$.

† The symbol $C_{n,p}$ denotes the binomial coefficient $n!/(p!)(n-p)!$.

证明．该定理的证明可类比于代数余子式展开公式的证明。然而，此处我们采用如下略有不同的方法。首先，我们证明：p \times p 阶顺序主子式与其互补子式之积恰好给出 A 中的 p ! (n - p) ! 项。事实上，若

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}^{\mathrm{c}}.$$

分别是该子式及其代数余子式中的任意项，则其乘积

$$A^{\mathrm{c}}\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} \triangleq (-1)^s A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}^{\mathrm{c}},$$

是行列式 det A , 的一项，因为 j _k (1 ≤ k ≤ p) 小于 j _r (p + 1 ≤ r ≤ n)。因此，数 t (j ^′) + t (j ^(′′)) 即为排列 j = (j □ , j □ , … , j _n) 中的逆序数，故其等于 t (j)。

由于一个 p × p 阶子式含有 p! 项，而与其对应的互补代数余子式含有 (n - p) ! 项，因此它们的乘积贡献了 det A 的 p ! (n - p) ! 项。

现在考虑 A 的选定行中任意一个子式

$$A\begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \end{pmatrix} = \begin{vmatrix} a_{21} & a_{22} & a_{24} \\ a_{31} & a_{32} & a_{34} \\ a_{51} & a_{52} & a_{54} \end{vmatrix},$$

设该子式位于 A 的选定行中。我们从它的第 i □ 行开始，依次将 A
的各行与其前面的行作对换，直至（5）式所示的子式最终位于 A 的前 p 行中。注意，这一过程共进行了 ∑ _(k=1) ^p i _k - ∑ _(k=1) ^p 次行对换。类似地，将（5）式所示子式移至 A 的第 p × p
个首主子矩阵的位置，需进行 ∑ _(k=1) ^p j _k ^ - ∑ _(k=1) ^p
次列对换。需注意：（5）式的互补子式并未改变；并且，依据本定理已证部分以及行列式的性质
3，该互补子式与原小矩阵的乘积提供了 det A 的 p ! (n - p) ! 项，其中 (- 1) ^s det A 的这些项由 s = ∑ _(k=1) ^p i _k + ∑ _(k=1) ^p j _k 给出。
(n - p) × (n - p)

现在只需将余子式替换为其对应的代数余子式即可。于是，从所选定的 p 行中选取的 C _(n,p)
个互异的子式，分别与其对应的余子式相乘，便得到 det A 的 p ! (n - p) !
个互异的项。因此，这些乘积之和恰好等于

$$A\begin{pmatrix}2 & 3 & 5\\ 1 & 2 & 4\end{pmatrix}^{c} = A\begin{pmatrix}1 & 4\\ 3 & 5\end{pmatrix} = \begin{vmatrix}a_{13} & a_{15}\\ a_{43} & a_{45}\end{vmatrix},$$

det A 中互异的项。

公式 (3) 的证明至此完成。公式 (4) 可由 (3) 并利用行列式的性质 2 直接推出。

习题 2．若 A□ 和 A□ 是方阵，证明

A □　A □

$$A^{c}\begin{pmatrix}2 & 3 & 5\\ 1 & 2 & 4\end{pmatrix} = (-1)^{s}\begin{vmatrix}a_{13} & a_{15}\\ a_{43} & a_{45}\end{vmatrix} = -\begin{vmatrix}a_{13} & a_{15}\\ a_{43} & a_{45}\end{vmatrix},$$

PROOF. The proof of this theorem can be arranged similarly to that of the cofactor expansion formula. However, we prefer here the following slightly different approach. First we prove that the product of the $p \times p$ leading principal minor with its complementary minor provides $p!(n-p)!$ elements of $\det A$. Indeed, if

$$(-1)^{t(j')}a_{1j_1}a_{2j_2}\cdots a_{pj_p} \qquad \text{and} \qquad (-1)^{t(j'')}a_{p+1,j_{p+1}}a_{p+2,j_{p+2}}\cdots a_{nj_n}$$

are any terms of the given minor and its cofactor, then the product

$$(-1)^{t(j')+t(j'')}a_{1j_1}a_{2j_2}\cdots a_{nj_n}$$

is a term of $\det A$, because j_k $(1 \le k \le p)$ is less than j_r $(p+1 \le r \le n)$. Therefore the number $t(j') + t(j'')$ is the number of inversions in the permutation $j = (j_1, j_2, \ldots, j_n)$, so it is equal to $t(j)$.

Since a $p \times p$ minor has $p!$ terms and the $(n-p) \times (n-p)$ corresponding complementary cofactor has $(n-p)!$ terms, their product provides $p!(n-p)!$ elements of $\det A$.

Now consider an arbitrary minor

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix} \tag{5}$$

lying in the chosen rows of A. Starting with its i_1th row, we successively interchange the rows of A with the preceding ones so that eventually the minor in (5) lies in the first p rows of A. Note that $\sum_{k=1}^{p} i_k - \sum_{k=1}^{p} k$ interchanges of rows are performed. Similarly, the shifting of (5) to the position of the $p \times p$ leading submatrix of A requires $\sum_{k=1}^{p} j_k - \sum_{k=1}^{p} k$ interchanges of columns. Observe that the complementary minor to (5) did not change and that, using the part of the theorem already proved and Property 3 of determinants, its product with the minor provides $p!(n-p)!$ terms of $(-1)^s \det A$, where $s = \sum_{k=1}^{p} i_k + \sum_{k=1}^{p} j_k$.

It suffices now to replace the complementary minor by the corresponding complementary cofactor. Thus, each of $C_{n,p}$ distinct products of minors lying in the chosen p rows with their complementary cofactors provides $p!(n-p)!$ distinct element of $\det A$. Consequently, the sum of these products gives exactly

$$C_{n,p}\,p!(n-p)! = n!$$

distinct terms of $\det A$.

The proof of (3) is completed. The formula in (4) follows from (3) by use of Property 2 of determinants. ■

Exercise 2. If A_1 and A_4 are square matrices, show that

$$\det\begin{bmatrix} A_1 & 0 \\ A_3 & A_4 \end{bmatrix} = (\det A_1)(\det A_4).$$

习题 3．证明：若方阵 A 具有分块三角形式，且主（分块）对角线上的各分块均为方阵 A _i (i = 1 , 2 , … , k)，则

$$(-1)^{t(j')}a_{1j_1}a_{2j_2}\cdots a_{pj_p} \qquad \text{and} \qquad (-1)^{t(j'')}a_{p+1,j_{p+1}}a_{p+2,j_{p+2}}\cdots a_{nj_n}$$

习题 4．证明：若 n × n 阶矩阵 A 具有如下形式

$$(-1)^{t(j')+t(j'')}a_{1j_1}a_{2j_2}\cdots a_{nj_n}$$

其中 A □ 与 A □ 分别为 p × p 阶与 (n - p) × (n - p) 阶矩阵，且 p < n ,，则

$$A\begin{pmatrix} i_1 & i_2 & \cdots & i_p \\ j_1 & j_2 & \cdots & j_p \end{pmatrix}$$

习题 5．计算 n \times n 阶矩阵 [- D _ B] . 的行列式，其中 B 是一个 n \times m 阶矩阵 (m < n)，而 D 是由 n \times n 阶单位矩阵 I _ { n } 删去下列各列后所得的矩阵：

$$C_{n,p}\,p!(n-p)! = n!$$

答案：

$$\det\begin{bmatrix} A_1 & 0 \\ A_3 & A_4 \end{bmatrix} = (\det A_1)(\det A_4).$$

其中

提示：注意到子矩阵 -D 仅有一个非零的 n - m , 阶子式，并利用公式 (4)。

习题 6．证明：若 m × n 矩阵 A 的所有 k 阶子式（(1 \leq k < \operatorname* { m i n } (m , n))）均为零，则 A 的任意阶数大于 k 的子式也必为零。

提示：对任意 k 行（或 k 列）应用拉普拉斯定理，将高阶子式展开为若干 k 阶子式与相应代数余子式的乘积之和。

2.5 比内–柯西公式

作为拉普拉斯定理的一个重要应用，我们证明如下关于两个矩阵乘积之行列式的结论。

定理 1（比内–柯西公式）．设 A 与 B 分别为 m \times n 矩阵与 n \times m 矩阵。若 m \leq n 且 C = A B ,，则

Exercise 3. Show that if a square matrix A is of a block-triangular form, with square blocks A_i $(i = 1, 2, \ldots, k)$ on the main (block) diagonal, then

$$\det A = \prod_{i=1}^{k} \det A_i.$$

Exercise 4. Prove that if the $n \times n$ matrix A is of the form

$$A = \begin{bmatrix} A_1 & A_2 \\ A_3 & 0 \end{bmatrix},$$

where A_2 and A_3 are $p \times p$ and $(n - p) \times (n - p)$ matrices, respectively, $p < n$, then

$$\det A = (-1)^{(n+1)p}(\det A_2)(\det A_3).$$

Exercise 5. Compute the determinant of the $n \times n$ matrix $[-D \quad B]$, where B is an $n \times m$ matrix $(m < n)$ and D is obtained from the $n \times n$ identity matrix I_n by striking out the columns

$$j_1, j_2, \ldots, j_m \quad (1 \le j_1 < j_2 < \cdots < j_m \le n).$$

Answer.

$$\det[-D \quad B] = (-1)^t B\begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix},$$

where

$$t = (n - m) + \left(\sum_{k=1}^{n} k + \sum_{k=1}^{m} j_k\right) - \sum_{k=1}^{n-m} k. \tag{6}$$

Hint. Observe that the submatrix $-D$ possesses only one nonzero minor of order $n - m$, and use (4).

Exercise 6. Show that if all minors of order k $(1 \le k < \min(m, n))$ of an $m \times n$ matrix A are equal to zero, then any minor of A of order greater than k is also zero.

Hint. Use Laplace's theorem for finding a larger minor by expansion along any k rows (or columns). □

2.5 The Binet–Cauchy Formula

As an important application of Laplace's theorem, we prove the following result concerning the determinant of a product of two matrices.

Theorem 1 (Binet–Cauchy formula). *Let A and B be $m \times n$ and $n \times m$ matrices, respectively. If $m \le n$ and $C = AB$, then*

$$\det C = \sum_{1 \le j_1 < j_2 < \cdots < j_m \le n} A\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} B\begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix}. \tag{1}$$

也就是说，乘积矩阵 AB 的行列式等于矩阵 A 的所有（最大阶数为 m 的）子式与矩阵 B 中相应阶数（即同为 m 阶）的对应子式的乘积之和。†

证明． 为证明该定理，我们首先构造一个分块三角形矩阵

(n + m) × (n + m)

其中 I _n 表示 n \times n 阶单位矩阵。

现利用习题 2.2.10 推出

$$\det A = \prod_{i=1}^{k} \det A_i .$$

从而，再由习题 2.4.4 可得

$$A = \begin{bmatrix} A_1 & A_2 \\ A_3 & 0 \end{bmatrix},$$

接下来，我们以另一种方式计算 \tilde { C }。由于

$$\det A = (-1)^{(n+1)p} (\det A_2)(\det A_3).$$

对至少一个 j _ { r } > n (1 \leq r \leq m) 成立，故根据拉普拉斯定理，可得

$$j_1, j_2, \ldots, j_m \quad (1 \le j_1 < j_2 < \cdots < j_m \le n).$$

回顾 \tilde { C } 的定义以及习题 2.4.5 的结论，不难看出

$$\det[-D \quad B] = (-1)^t B \begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix},$$

$$t = (n - m) + \left(\sum_{k=1}^{n} k + \sum_{k=1}^{m} j_k \right) - \sum_{k=1}^{n-m} k.$$

其中 \begin{array} { r } { s = \sum _ { k = 1 } ^ { m } k + \sum _ { k = 1 } ^ { m } j _ { k } } \end{array}，且 t 由式 (2.4.6) 给出。‡ 稍作计算可得

$$\det C = \sum_{1 \le j_1 < j_2 < \cdots < j_m \le n} A\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} B\begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix}.$$

因此

该值恒为偶数。

That is, the determinant of the product AB is equal to the sum of the products of all possible minors of (the maximal) order m of A with corresponding minors of B of the same order.†

PROOF. To prove the theorem, we first construct a block-triangular $(n + m) \times (n + m)$ matrix

$$\tilde{C} = \begin{bmatrix} A & 0 \\ -I_n & B \end{bmatrix},$$

where I_n denotes the $n \times n$ identity matrix. Now use Exercise 2.2.10 to deduce

$$\det \tilde{C} = \det \begin{bmatrix} A & AB \\ -I_n & 0 \end{bmatrix} = \det \begin{bmatrix} A & C \\ -I_n & 0 \end{bmatrix}$$

and hence, by Exercise 2.4.4,

$$\det \tilde{C} = (-1)^{(m+n+1)m}(\det C)(-1)^n = (-1)^{(m+n)(m+1)} \det C. \tag{2}$$

Now we evaluate $\det \tilde{C}$ in a different way. Since

$$\tilde{C}\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} = 0$$

for at least one $j_r > n$ $(1 \leq r \leq m)$, it follows, using Laplace's theorem, that

$$\det \tilde{C} = \sum_{1 \leq j_1 < j_2 \cdots < j_m \leq n} A\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} \tilde{C}^{c}\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}. \tag{3}$$

Recalling the definition of $\tilde{C}$ and the result of Exercise 2.4.5, it is not difficult to see that

$$\tilde{C}^{c}\begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} = (-1)^s(-1)^t B\begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix}, \tag{4}$$

where $s = \sum_{k=1}^{m} k + \sum_{k=1}^{m} j_k$ and t is given by Eq. (2.4.6).‡ A little calculation gives

$$s + t = n(m + 1) + 2\sum_{k=1}^{m} j_k,$$

and so

$$(s + t) - (m + n)(m + 1) = 2\sum_{k=1}^{m} j_k - m(m + 1),$$

which is always even.

† For $m > n$ see Proposition 3.8.4.

‡ Note that in the case $n = m$, we have $t = 0$.

将 (4) 式代入 (3) 式，并与 (2) 式比较，即完成本定理的证明。■

在 (1) 式中令 m = n，可得如下有用结论：

推论 1．两个方阵 A 与 B 之乘积的行列式，等于各因子行列式的乘积：

$$\tilde{C} = \begin{bmatrix} A & 0 \\ -I_n & B \end{bmatrix},$$

例 1．设

$$\det \tilde{C} = \det \begin{bmatrix} A & AB \\ -I_n & 0 \end{bmatrix} = \det \begin{bmatrix} A & C \\ -I_n & 0 \end{bmatrix}$$

则显然有

$$\det \tilde{C} = (-1)^{(m+n+1)m} (\det C)(-1)^n = (-1)^{(m+n)(m+1)} \det C.$$

且 C = 2。若改用比内–柯西公式（Binet–Cauchy formula）计算，亦得相同结果：

$$\tilde{C} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} = 0$$

习题 2．验证：对任意 k × n 阶矩阵 A，有 k ≤ n _:。

$$\det \tilde{C} = \sum_{1 \le j_1 < j_2 \cdots < j_m \le n} A \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} \tilde{C}^{\mathrm{c}} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix}.$$

习题 3．应用比内–柯西公式（Binet–Cauchy formula），证明

$$\tilde{C}^{\mathrm{c}} \begin{pmatrix} 1 & 2 & \cdots & m \\ j_1 & j_2 & \cdots & j_m \end{pmatrix} = (-1)^s (-1)^t B \begin{pmatrix} j_1 & j_2 & \cdots & j_m \\ 1 & 2 & \cdots & m \end{pmatrix},$$

Substituting (4) in (3) and comparing with (2), we complete the proof of the theorem. ■

By putting $m = n$ in (1), the following useful fact is obtained:

Corollary 1. *The determinant of the product of two square matrices A and B is the product of the determinants of the factors*:

$$\det AB = (\det A)(\det B). \tag{5}$$

Example 1. Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & -1 \\ -2 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then obviously

$$C = AB = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

and $\det C = 2$. The same result is obtained by using the Binet–Cauchy formula:

$$\begin{aligned} \det C &= \sum_{1 \le j_1 < j_2 \le 3} A\begin{pmatrix} 1 & 2 \\ j_1 & j_2 \end{pmatrix} B\begin{pmatrix} j_1 & j_2 \\ 1 & 2 \end{pmatrix} \\ &= A\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} B\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} + A\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} B\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix} + A\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} B\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \\ &= \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} \begin{vmatrix} -1 & -1 \\ -2 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} \begin{vmatrix} -1 & -1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} \\ &= 2. \end{aligned}$$

Exercise 2. Check that for any $k \times n$ matrix A, $k \le n$,

$$\det AA^* = \sum_{1 \le j_1 < j_2 < \cdots < j_k \le n} \left| A\begin{pmatrix} 1 & 2 & \cdots & k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \right|^2 \ge 0.$$

Exercise 3. Applying the Binet–Cauchy formula, prove that

$$\begin{vmatrix} \sum_{i=1}^{n} a_i c_i & \sum_{i=1}^{n} a_i d_i \\ \sum_{i=1}^{n} b_i c_i & \sum_{i=1}^{n} b_i d_i \end{vmatrix} = \sum_{1 \le j < k \le n} \begin{vmatrix} a_j & a_k \\ b_j & b_k \end{vmatrix} \begin{vmatrix} c_j & c_k \\ d_j & d_k \end{vmatrix}$$

并由此导出柯西恒等式（Cauchy identity）：

$$\det AB = (\det A)(\det B).$$

提示：利用如下矩阵

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & -1 \\ -2 & 0 \\ 1 & 1 \end{bmatrix}.$$

习题 4．证明实数域上的柯西–施瓦茨不等式：

$$C = AB = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$$

†

提示：利用习题 3 的结论。

习题 5．验证：任意有限个同阶方阵之乘积的行列式，等于各因子行列式的乘积。

2.6 伴随矩阵与逆矩阵

设 A = [a _(ij)] _(i,j=1) ^n 为任一 n × n 矩阵；如第 2.4 节所定义，记 A _(ij) = A ^c i j 为元素 a _(ij) (1 ≤ i , j ≤ n) 的代数余子式。矩阵 A , 的伴随矩阵（adjoint matrix），记作 adjA ,，定义为 A
的所有代数余子式构成的矩阵的转置。因此

$$\begin{aligned} \det C &= \sum_{1 \le j_1 < j_2 \le 3} A\begin{pmatrix} 1 & 2 \\ j_1 & j_2 \end{pmatrix} B\begin{pmatrix} j_1 & j_2 \\ 1 & 2 \end{pmatrix} \\ &= A\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} B\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} + A\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} B\begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix} + A\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} B\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} \\ &= \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} \begin{vmatrix} -1 & -1 \\ -2 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} \begin{vmatrix} -1 & -1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} -2 & 0 \\ 1 & 1 \end{vmatrix} \\ &= 2. \end{aligned}$$

习题 1．验证：若

$$\det AA^* = \sum_{1 \le j_1 < j_2 < \cdots < j_k \le n} \left| A \begin{pmatrix} 1 & 2 & \cdots & k \\ j_1 & j_2 & \cdots & j_k \end{pmatrix} \right|^2 \ge 0.$$

and deduce the *Cauchy identity*:

$$\left(\sum_{i=1}^{n} a_i c_i\right)\left(\sum_{i=1}^{n} b_i d_i\right) - \left(\sum_{i=1}^{n} a_i d_i\right)\left(\sum_{i=1}^{n} b_i c_i\right) = \sum_{1 \le j < k \le n} (a_j b_k - a_k b_j)(c_j d_k - c_k d_j).$$

Hint. Use the matrices

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \end{bmatrix}, \qquad B = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \\ d_1 & d_2 & \cdots & d_n \end{bmatrix}^{\mathrm{T}}.$$

Exercise 4. Prove the Cauchy–Schwarz inequality,

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right),$$

in real numbers.†

Hint. Use Exercise 3.

Exercise 5. Check that the determinant of the product of any (finite) number of square matrices of the same order is equal to the product of the determinants of the factors. □

2.6 Adjoint and Inverse Matrices

Let $A = [a_{ij}]_{i,j=1}^{n}$ be any $n \times n$ matrix and, as defined in Section 2.4, let $A_{ij} = A^{\mathrm{c}}\binom{i}{j}$ be the cofactor of a_{ij} $(1 \le i, j \le n)$. The *adjoint* of A, written adj A, is defined to be the transposed matrix of cofactors of A. Thus

$$\operatorname{adj} A \triangleq ([A_{ij}]_{i,j=1}^{n})^{\mathrm{T}}.$$

Exercise 1. Check that if

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix},$$

† For a straightforward proof of this inequality (in a more general context), see Theorem 3.11.1.

则

$$\left(\sum_{i=1}^{n} a_i c_i\right)\left(\sum_{i=1}^{n} b_i d_i\right) - \left(\sum_{i=1}^{n} a_i d_i\right)\left(\sum_{i=1}^{n} b_i c_i\right)$$

$$= \sum_{1 \le j < k \le n} (a_j b_k - a_k b_j)(c_j d_k - c_k d_j).$$

伴随矩阵的一些性质可直接由其定义推出。

习题 2．验证：对任意 n × n 矩阵 A 以及任意 k ∈ F

$$A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \end{bmatrix}, \qquad B = \begin{bmatrix} c_1 & c_2 & \cdots & c_n \\ d_1 & d_2 & \cdots & d_n \end{bmatrix}^{\mathrm{T}}.$$

以下这一引人注目的结论及其推论，正是人们关注矩阵伴随矩阵（adjoint）的主要原因。

定理 1. 对任意 n × n 阶矩阵 A，有

$$\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le \left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right),$$

证明. 利用行展开公式（2.3.1）及习题 2.3.3
的结果计算左边的乘积，可得一个矩阵：其主对角线上每个位置的元素均为
A，其余位置均为零，即该乘积等于 (A)I。
习题 3. 证明：

$$\operatorname{adj} A \triangleq ([A_{ij}]_{i,j=1}^{n})^{\mathrm{T}}.$$

定理 1 具有一些重要推论。例如，若 A 是 n × n 阶矩阵且 A \neq 0 ,，则

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix},$$

事实上，利用行列式性质 A B = (\operatorname* { d e t } A) (\operatorname* { d e t } B)、式 (1)
以及对角矩阵行列式的已知结果，即可推出该结论。

A 是否为零，是方阵的一项极为关键的性质，因而值得专门定义：一个方阵 A 称为奇异的（singular）或非奇异的（nonsingular），当且仅当 $A = 0$ 或 $A \neq 0$。

then

$$\operatorname{adj} A = \begin{bmatrix} \begin{vmatrix} 2 & 0 \\ -1 & 3 \end{vmatrix} & -\begin{vmatrix} 0 & 0 \\ 1 & 3 \end{vmatrix} & \begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix} \\ -\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} 6 & -2 & 2 \\ 0 & 7 & 0 \\ -2 & 3 & 4 \end{bmatrix}. \quad \square$$

Some properties of adjoint matrices follow immediately from the definition.

Exercise 2. Check that for any $n \times n$ matrix A and any $k \in \mathscr{F}$,

$$\operatorname{adj}(A^{\mathrm{T}}) = (\operatorname{adj} A)^{\mathrm{T}},$$

$$\operatorname{adj}(A^*) = (\operatorname{adj} A)^*,$$

$$\operatorname{adj} I = I,$$

$$\operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A. \quad \square$$

The following striking result, and its consequences, are the main reason for interest in the adjoint of a matrix.

Theorem 1. *For any $n \times n$ matrix A,*

$$A(\operatorname{adj} A) = (\det A)I. \tag{1}$$

PROOF. Computing the product on the left by use of the row expansion formula (2.3.1) and the result of Exercise 2.3.3, we obtain a matrix with the number $\det A$ in each position on its main diagonal and zeros elsewhere, that is, $(\det A)I$. ■

Exercise 3. Prove that

$$(\operatorname{adj} A)A = (\det A)I. \quad \square \tag{2}$$

Theorem 1 has some important consequences. For example, if A is $n \times n$ and $\det A \neq 0$, then

$$\det(\operatorname{adj} A) = (\det A)^{n-1}. \tag{3}$$

In fact, using the property $\det AB = (\det A)(\det B)$ with Eq. (1) and the known determinant of the diagonal matrix, the result follows.

The vanishing or nonvanishing of $\det A$ turns out to be a vitally important property of a square matrix and warrants a special definition: A square matrix A is said to be *singular* or *nonsingular* according to whether $\det A$ is zero or nonzero.

习题 4. 验证下列矩阵

$$\operatorname{adj} A = \begin{bmatrix} \begin{vmatrix} 2 & 0 \\ -1 & 3 \end{vmatrix} & -\begin{vmatrix} 0 & 0 \\ 1 & 3 \end{vmatrix} & \begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix} \\ -\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} 6 & -2 & 2 \\ 0 & 7 & 0 \\ -2 & 3 & 4 \end{bmatrix}. \quad \square$$

分别为奇异矩阵与非奇异矩阵。

设 A 为一个 n × n 矩阵，若 n × n 矩阵 B 满足

$$\operatorname{adj}(A^{\mathrm{T}}) = (\operatorname{adj} A)^{\mathrm{T}},$$

$$\operatorname{adj}(A^{*}) = (\operatorname{adj} A)^{*},$$

$$\operatorname{adj} I = I,$$

$$\operatorname{adj}(kA) = k^{n-1} \operatorname{adj} A. \quad \square$$

则称 B 为 A 的一个逆矩阵；此时记作 B = A □¹。此外，当 A 存在逆矩阵时，称 A 是可逆的。我们将在本节中证明：一个方阵可逆当且仅当它是非奇异的（本节定理 2）。

习题 5．验证下列矩阵

$$A(\operatorname{adj} A) = (\det A)I.$$

互为逆矩阵。

习题

6．证明：一个方阵若是上（下）三角矩阵，则它可逆当且仅当其主对角线上无零元。在此条件下，进一步验证其逆矩阵仍为同类型的三角矩阵。

习题 7．验证：对合矩阵（即满足 A² = I 的矩阵）——且仅此类矩阵——是其自身的逆矩阵。

习题 8．验证：若 A □¹ 存在，则

$$(\operatorname{adj} A)A = (\det A)I. \quad \square$$

逆矩阵的概念源于实数域（或复数域）这一人们所熟知的数域。事实上，若 a ≠ 0 ,，则可将商 b = 1 / a
视为数 ±b a , 的逆元（或倒数），且该倒数 b 满足 a b =□ba = 1，正如式 (4) 所示。当然，我们亦可记作 b = a □¹。然而，矩阵乘法的非交换性，尤其是如第 1.3
节所述，矩阵缺乏某些数所具有的性质，导致逆矩阵的代数性质与数的倒数之间存在实质性差异。
例如，如下一习题所示，并非所有非零矩阵都存在逆矩阵。

Exercise 4. Confirm that the matrices

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & -4 \end{bmatrix}$$

are, respectively, singular and nonsingular. □

An $n \times n$ matrix B is referred to as an *inverse* of the $n \times n$ matrix A if

$$AB = BA = I, \tag{4}$$

and when this is the case, we write $B = A^{-1}$. Also, when A has an inverse, A is said to be *invertible*. We shall see that a square matrix is invertible if and only if it is nonsingular (Theorem 2 of this section).

Exercise 5. Check that the matrices

$$A = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 1 & -2 & 2 \end{bmatrix}$$

are mutually inverse matrices.

Exercise 6. Show that a square upper- (lower-) triangular matrix is invertible if and only if there are no zeros on the main diagonal. If this is the case, verify that the inverse is a matrix of the same kind.

Exercise 7. Check that the involutory matrices, and only they, are their own inverses.

Exercise 8. Check that if A^{-1} exists, then

$$(A^{-1})^{\mathrm{T}} = (A^{\mathrm{T}})^{-1}. \quad \square$$

The notion of an inverse matrix has its origin in the familiar field of the real (or complex) numbers. Indeed, if $a \neq 0$, we may consider the quotient $b = 1/a$ as the inverse (or reciprocal) of the number a, and b satisfies $ab = ba = 1$, as in Eq. (4). We also write $b = a^{-1}$, of course. However, the noncommutativity of matrix multiplication and, especially, the lack of some of the properties of numbers, as mentioned in Section 1.3, yield substantial differences between the algebraic properties of inverse matrices and reciprocals of numbers.

For instance, as the next exercise shows, not all nonzero matrices have an inverse.

习题 9．验证：若

$$A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & -4 \end{bmatrix}$$

则不存在 3 × 3 矩阵 B，使得

$$AB = BA = I,$$

应当强调的是，鉴于矩阵乘法不满足交换律，若 B 是 A 的逆矩阵，则根据定义，B 必须同时是 A 的左逆，即满足 (B A = I)，又必须是 A 的右逆，即满足 r i g h t□(A B = I)。不过，我们很快将看到：对任一 n 阶方阵而言，只要它存在单侧逆（左逆或右逆），该单侧逆必自动成为双侧逆。

此外还须指出，定义 (4) 平凡地给出了矩阵可逆的一个必要条件：即若一方阵 A 至少存在一个单侧逆，则 A 必为非奇异矩阵，亦即其行列式 det□A ≠ 0 .。事实上，若例如 A B = I ,，则

$$A = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ 1 & -2 & 2 \end{bmatrix}$$

并且显然，det A ≠ 0 .。事实上，条件 det A ≠ 0 对于矩阵 A 存在逆矩阵而言也是充分的。

定理 2. 一个方阵可逆当且仅当它是非奇异的。

证明. 若 det A ≠ 0 ,，则公式 (1) 和 (2) 表明，矩阵

$$(A^{-1})^{\mathrm{T}} = (A^{\mathrm{T}})^{-1}. \quad \square$$

满足条件 (4)，因此即为 A 的逆矩阵。■

在 det A ≠ 0 的情形下，逆矩阵的存在性可直接推出其唯一性。为说明这一点，首先验证右逆的唯一性：即若 A B □ = I 且 A B □ = I ,，则必有 ±b B □ = ±b B □。事实上，由 A B □ = A B □ 可得 A (B □ - B □) = 0 ,；又因 A 存在右逆，故 det A ≠ 0。应用定理 ^(2,)，我们可推出 A □¹ 的存在性。在等式两端左乘 A □¹，即得

从而得到所要求的等式 B □ = B □。

类似地，可证非奇异矩阵的左逆亦唯一。因此，若 det A ≠ 0 ,，则其左、右单侧逆均唯一；而且由于二者均由公式 (5) 给出，故彼此相等。

Exercise 9. Check that if

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

then there is no 3×3 matrix B such that

$$AB = I. \quad \square$$

It should be emphasized that, in view of the lack of commutativity of matrix multiplication, if B is an inverse of A the definition requires that it must simultaneously be a *left* inverse ($BA = I$) and a *right* inverse ($AB = I$). However, it will be seen shortly that a one-sided inverse for a *square* matrix is necessarily two-sided.

Furthermore, it should be noted that the definition (4) provides trivially a necessary condition for a matrix to be invertible. Namely, if a square matrix A has even a one-sided inverse, then A is nonsingular, that is, $\det A \neq 0$. Indeed, if for instance $AB = I$, then

$$\det AB = (\det A)(\det B) = 1$$

and, obviously, $\det A \neq 0$. It turns out that the condition $\det A \neq 0$ is also sufficient for A to have an inverse.

Theorem 2. *A square matrix is invertible if and only if it is nonsingular.*

PROOF. If $\det A \neq 0$, then formulas (1) and (2) show that the matrix

$$B = \frac{1}{\det A} \operatorname{adj} A \tag{5}$$

satisfies conditions (4) and, therefore, is an inverse of A. ■

The existence of an inverse in the case $\det A \neq 0$ easily implies its uniqueness. To see this, first check that the right inverse is unique, that is, if $AB_1 = I$ and $AB_2 = I$, then $B_1 = B_2$. Indeed, $AB_1 = AB_2$ implies $A(B_1 - B_2) = 0$, and since A has a right inverse, then $\det A \neq 0$. Applying Theorem 2, we deduce the existence of A^{-1}. Multiplying by A^{-1} from the left we have

$$A^{-1}A(B_1 - B_2) = 0,$$

and the required equality $B_1 = B_2$ is obtained.

Similarly, the uniqueness of the left inverse of a nonsingular matrix is established. Hence if $\det A \neq 0$, then both one-sided inverses are unique. Furthermore, since they are both given by Eq. (5), they are equal.

习题 10. 利用公式 (5) 验证：

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

利用公式 (5)
对大型矩阵进行数值计算，通常在计算时间上是臭名昭著地昂贵。其他计算逆矩阵的方法将在习题 2.7.4 中给出，这些方法为通用算法提供了更优的基础。

习题 11. 验证：若 A 是非奇异矩阵，则

$$AB = I. \quad \square$$

习题 12．确认：同阶方阵的任意有限个乘积可逆，当且仅当每个因子均可逆。

习题 13．证明：

$$\det AB = (\det A)(\det B) = 1$$

其中 A 与 B 是同型可逆矩阵。

习题 14．证明：若 A 与 B 是同型非奇异矩阵，且 A = B，则存在一个行列式为 1 的矩阵 C，使得
A = B C

习题 15．验证：若 A 是非奇异矩阵，D 是方阵（A 与 D 的阶数可以不同），则

$$B = \frac{1}{\det A} \operatorname{adj} A$$

提示：利用如下关系式

$$A^{-1}A(B_1 - B_2) = 0,$$

以及习题 2.4.2 和 2.5.5。（值得注意的是，式 (6) 与一个 2 × 2
矩阵的行列式公式之间存在类比关系。另需注意：当 B、C 或 D 可逆时，可类似地轻易导出与式 (7) 形式相近的公式。）矩阵 D - C A □¹ B 被称为 A 在 [A B C D] . 中的舒尔补（Schur complement）。

习题 16. 若矩阵 A ∈ R ^(n×n) 满足 A ^T A = I .，则称其为正交矩阵。试证明：

(a) 正交矩阵 A 可逆，且 A = \pm 1，同时 \pmb { A } ^ { - 1 } = \pmb { A } ^ { \mathnormal { \tilde { \mathbf { I } } } } };

Exercise 10. Check, using Eq. (5), that

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix}^{-1} = \tfrac{1}{14}\begin{bmatrix} 6 & -2 & 2 \\ 0 & 7 & 0 \\ -2 & 3 & 4 \end{bmatrix}. \quad \square$$

Numerical calculations with Eq. (5) for large matrices are, in general, notoriously expensive in computing time. Other methods for calculating the inverse will be presented in Exercise 2.7.4 and provide a better basis for general-purpose algorithms.

Exercise 11. Check that if A is nonsingular, then

$$\det(A^{-1}) = (\det A)^{-1}.$$

Exercise 12. Confirm that the product of any number of square matrices of the same order is invertible if and only if each factor is invertible.

Exercise 13. Prove that

$$(AB)^{-1} = B^{-1}A^{-1},$$

provided A and B are invertible matrices of the same size.

Exercise 14. Show that if A and B are nonsingular matrices of the same size and $\det A = \det B$, then there exists a matrix C with determinant 1 such that $A = BC$.

Exercise 15. Check that if A is a nonsingular matrix and D is square (A and D may be of different sizes), then

$$\det\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D - CA^{-1}B)\det A. \tag{6}$$

Hint. Use the relation

$$\begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix}\begin{bmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{bmatrix}\begin{bmatrix} I & A^{-1}B \\ 0 & I \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \tag{7}$$

as well as Exercises 2.4.2 and 2.5.5. (It is interesting to observe the analogy between Eq. (6) and the formula for the determinant of a 2×2 matrix. Note also that formulas similar to Eq. (7) in the cases when B, C, or D are invertible can be easily derived.) The matrix $D - CA^{-1}B$ is known as the *Schur complement* of A in $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

Exercise 16. A matrix $A \in \mathbb{R}^{n \times n}$ is said to be *orthogonal* if $A^{T}A = I$. Show that:

(a) An orthogonal matrix A is invertible, $\det A = \pm 1$, and $A^{-1} = A^{T}$.

(b) I f n = 2 ,，所有正交矩阵必具有以下两种形式之一：

$$\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix}^{-1} = \frac{1}{14}\begin{bmatrix} 6 & -2 & 2 \\ 0 & 7 & 0 \\ -2 & 3 & 4 \end{bmatrix}. \quad \square$$

对某个角度 θ 成立。

习题 17．若矩阵 U ∈ C ^(n×n) 满足 U ^* U = I，则称其为酉矩阵（unitary matrix）。试证明：此时 U 可逆，且 |\det\,

$$\det(A^{-1}) = (\det A)^{-1}.$$

\begin{array}{c} E ^ { (1) } = \begin{array} { c } { { \left[\begin{array} { l } { { 1 } } \\ { { \ddots } } \\ { { 1 } } \end{array} \right] } } \\ { { \begin{array} { r l } { { (i _ { 1 }) } } \\ { { 0 } } \end{array} \quad \cdots \quad 1 } } \\ { { \begin{array} { r l } { { 1 } } \\ { { \vdots } } \\ { { 1 } } \end{array} \quad \ddots \quad \vdots } } \\ { { \begin{array} { r l } { { 1 } } \\ { { (i _ { 2 }) } } \\ { { 1 } } \end{array} \quad \cdots \quad 0 } } \end{array} ， \\ { { \begin{array} { r l } { { \begin{array} { r l } { { 1 } } \\ { { \ddots } } \\ { { 1 } } \end{array} \quad \cdots \quad 0 } } \\ { { \begin{array} { r l } { { 1 } } \\ { { \ddots } } \\ { { 1 } } \end{array} \quad \ddots } } \\ { { \begin{array} { r l } { { 1 } } \end{array} \quad } } \end{array} } } \end{array}\tag{1}

(b) If $n = 2$, all orthogonal matrices have one of the two forms,

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix},$$

for some angle θ.

Exercise 17. A matrix $U \in \mathbb{C}^{n\times n}$ is said to be *unitary* if $U^*U = I$. Show that, in this case, U is invertible, $|\det U| = 1$ and $U^{-1} = U^*$. □

2.7 Elementary Operations on Matrices

Properties 1 through 5 of determinants indicate a close relationship between the determinant of a square matrix A and matrices obtained from A by the following operations:

(1) interchanging two rows (or columns) in A;
(2) multiplying all elements of a row (or column) of A by some nonzero number;
(3) adding to any row (or column) of A any other row (or column) of A multiplied by a nonzero number.

The operations 1 through 3 listed above are referred to as *elementary row* (respectively, *column*) *operations of types* 1, 2, and 3 respectively.

Now observe that these manipulations of the rows and columns of a matrix A can be achieved by pre- and postmultiplication (respectively) of A by appropriate matrices. In particular, the interchange of rows i_1 and i_2 ($i_1 < i_2$) of A can be performed by multiplying A from the left by the $n \times n$ matrix

$$E^{(1)} = \begin{matrix} \\ \\ \\ (i_1) \\ \\ \\ (i_2) \\ \\ \\ \end{matrix} \begin{bmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & 0 & \cdots & 1 & & \\ & & & & 1 & & & \\ & & & \vdots & & \vdots & & \\ & & & & & 1 & & \\ & & & 1 & \cdots & 0 & & \\ & & & & & & 1 & \\ & & & & & & & \ddots \\ & & & & & & & & 1 \end{bmatrix}, \tag{1}$$

通过交换单位矩阵的第 i □ 行与第 i □ 行所得。因此，矩阵 E □¹□ A 的各行与矩阵 A 相同，仅第 i □ 行与第 i □ 行互换了位置。

此外，将矩阵 A 的第 i 行乘以一个非零数 k 所产生的效果，可通过计算乘积 E □²□ A 实现，其中

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \qquad \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix},$$

最后，将矩阵 A 的第 i □ 行的 k 倍加到第 i □ 行上，等价于用某个矩阵左乘 A。

$$E^{(1)} = \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & & 0 & \cdots & 1 & \\ & & & & 1 & & \\ & & & \vdots & \ddots & \vdots & \\ & & & & & 1 & \\ & & & 1 & \cdots & 0 & \\ & & & & & & 1 \\ & & & & & & & \ddots \\ & & & & & & & & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ (i_1) \\ \\ \\ \\ (i_2) \\ \\ \\ \\ \end{matrix},$$

取决于 i _ { 1 } < i _ { 2 } 还是 i _ { 1 } > i _ { 2 }。再次注意：对矩阵 A 所施行的运算，若作用于单位矩阵 I 上，则生成矩阵 E ^ { (3) }。

类似地，用适当的矩阵 E ^ { (1) }, E ^ { (2) }、E ^ { (3) } 右乘 A，将导致其列向量发生相应的变动。

矩阵 E ^ { (1) }, E ^ { (2) }、E ^ { (3) } 分别称为第一类、第二类和第三类初等矩阵。

行列式的性质 1、3 和 5 现可重述如下：

习题 1．验证

obtained by interchanging rows i_1 and i_2 of the identity matrix. Thus, the matrix $E^{(1)}A$ has the same rows as A except that rows i_1 and i_2 are interchanged.

Furthermore, the effect of multiplying the ith row of A by a nonzero number k can be achieved by forming the product $E^{(2)}A$, where

$$E^{(2)} = (i)\begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & k & & \\ & & & & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix}. \tag{2}$$

Finally, adding k times row i_2 to row i_1 of A is equivalent to multiplication of A from the left by the matrix

$$E^{(3)} = \begin{matrix} \\ (i_1) \\ \\ (i_2) \\ \\ \end{matrix}\begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & \cdots & k & & \\ & & & 1 & \vdots & & \\ & & & & \ddots & & \\ & & & & 1 & & \\ & & & & & 1 & \\ & & & & & & \ddots \\ & & & & & & & 1 \end{bmatrix} \quad \text{or} \quad E^{(3)} = \begin{matrix} \\ (i_2) \\ \\ (i_1) \\ \\ \end{matrix}\begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \vdots & \ddots & \\ & & k & \cdots 1 & \\ & & & & \ddots \\ & & & & & 1 \end{bmatrix}, \tag{3}$$

depending on whether $i_1 < i_2$ or $i_1 > i_2$. Again note that the operation to be performed on A is applied to I in order to produce $E^{(3)}$.

Similarly, the multiplication of A on the *right* by the appropriate matrices $E^{(1)}$, $E^{(2)}$, or $E^{(3)}$ leads to analogous changes in columns.

The matrices $E^{(1)}$, $E^{(2)}$, and $E^{(3)}$ are called *elementary matrices* of *types* 1, 2, and 3, respectively.

Properties 1, 3, and 5 of determinants may now be rewritten as follows:

$$\begin{aligned} \det E^{(1)}A &= \det AE^{(1)} = -\det A, \\ \det E^{(2)}A &= \det AE^{(2)} = k \det A, \\ \det E^{(3)}A &= \det AE^{(3)} = \det A. \end{aligned} \tag{4}$$

Exercise 1. Check that

$$\det E^{(1)} = -1, \qquad \det E^{(2)} = k, \qquad \det E^{(3)} = 1.$$

习题 2．验证初等矩阵的逆矩阵仍是同类型的初等矩阵。

初等变换的应用可将一个一般的 m × n

矩阵化为更简单的形式，该简化矩阵尤其含有比原矩阵更多的零元。这一方法称为高斯约化（或消元）过程，如下所示。事实上，我们在例 2.2.4 中计算行列式时已使用过该过程。

例 3．考虑矩阵

$$E^{(2)} = (i)\begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & k & & \\ & & & & 1 & \\ & & & & & \ddots \\ & & & & & & 1 \end{bmatrix}.$$

将第二行乘以 - 1 2 ,，得到

$$E^{(3)} = \begin{matrix} (i_1) \\ \\ (i_2) \\ \\ \\ \\ \end{matrix}\begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & \cdots & k & & \\ & & & 1 & \vdots & & \\ & & & & \ddots & & \\ & & & & 1 & & \\ & & & & & 1 & \\ & & & & & & \ddots \\ & & & & & & & 1 \end{bmatrix} \quad \text{or} \quad E^{(3)} = \begin{matrix} \\ \\ (i_2) \\ \\ (i_1) \\ \\ \\ \end{matrix}\begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & & & & \\ & & \vdots & \ddots & & & \\ & & k & \cdots & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix},$$

其中 E □ □²□ 是一个 3 \times 3 阶的第二类初等矩阵，且满足 k = - 1 2 , i = 2 .。

将第二行乘以 -4 后加到第三行，可得

$$\det E^{(1)}A = \det AE^{(1)} = -\det A,$$

$$\det E^{(2)}A = \det AE^{(2)} = k \det A,$$

$$\det E^{(3)}A = \det AE^{(3)} = \det A.$$

其中 E □ □³□ 是一个 3 \times 3 阶的第三类初等矩阵，且满足 k = - 4 ,、i □ = 3 , i □ = 2。

交换前两行得到

$$\det E^{(1)} = -1, \qquad \det E^{(2)} = k, \qquad \det E^{(3)} = 1.$$

且所得矩阵的第一列除一个元素等于 1 外，其余元素均为零。

再利用一个类型为 3 的 3 × 3 矩阵（其中 k = 2 , i □ = 3 , i □ = 2），我们得到

显然，A □ 是给定矩阵 A 在仅施行初等行变换的前提下所能达到的最简形式。该形式称为 A 的“行最简形”（reduced row-echelon form），在求解线性方程组中被广泛应用（参见第 2.9 节）。

若进一步施行适当的初等列变换，则不难看出，矩阵 A 还可进一步化为对角形：

其中 E □ □³□ , E □ □³□ , E □ □³□ 表示如下类型为 (3) 的矩阵：

综上所述，所有上述化简过程可归纳为

或

其中矩阵 P 与 Q 均为初等矩阵的乘积，因此均为非奇异矩阵。□

将上述消元过程应用于一般的 m × n 矩阵，可得到如下重要结论；该结论不加形式证明地给出。

定理 1. 任一非零的 m × n 矩阵均可通过初等行变换化为行最简形（reduced row echelon form），该形式具有如下定义性质：

(1) 所有零行均位于矩阵的底部；

(2) 每个非零行中从左起第一个非零元素为 1（称为“首元”或“主元”）。

（3）对于矩阵 j = 2 , 3 , \dots , m，第 j 行的首元（若存在）位于第 j - 1 行首元的右侧；

（4）任何包含首元的列，其其余所有元素均为零。

Exercise 2. Check that the inverses of the elementary matrices are also elementary of the same type. □

The application of elementary operations admits the reduction of a general $m \times n$ matrix to a simpler one, which contains, in particular, more zeros than the original matrix. This approach is known as the *Gauss reduction* (or *elimination*) *process* and is illustrated below. In fact, it was used in Example 2.2.4 for computing a determinant.

Example 3. Consider the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ -2 & 0 & 0 & 6 \\ 4 & -2 & -4 & -10 \end{bmatrix}.$$

Multiplying the second row by $-\frac{1}{2}$, we obtain

$$A_1 = E_1^{(2)}A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & -3 \\ 4 & -2 & -4 & -10 \end{bmatrix},$$

where $E_1^{(2)}$ is a 3×3 matrix of the type 2 with $k = -\frac{1}{2}$, $i = 2$.

Multiplying the second row by -4 and adding it to the third, we deduce

$$A_2 = E_2^{(3)}A_1 = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & -3 \\ 0 & -2 & -4 & 2 \end{bmatrix},$$

where $E_2^{(3)}$ is a 3×3 elementary matrix of the third type with $k = -4$, $i_1 = 3, i_2 = 2$.

The interchange of the first two rows provides

$$A_3 = E_3^{(1)}A_2 = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & -2 & -4 & 2 \end{bmatrix},$$

and the first column of the resulting matrix consists only of zeros except one element equal to 1.

Employing a 3×3 matrix of type 3, where $k = 2, i_1 = 3, i_2 = 2$, we obtain

$$A_4 = E_4^{(3)}A_3 = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{5}$$

and it is clear that A_4 is the simplest form of the given matrix A that can be achieved by row elementary operations. This is the "reduced row-echelon

以下矩阵均具有行最简形（即简化行阶梯形）：

$$A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ -2 & 0 & 0 & 6 \\ 4 & -2 & -4 & -10 \end{bmatrix}.$$

{ \begin{array} { r l } { { \underline { { \left[1 \quad 0 \quad - 3 \right] } } } , \quad { \underline { { \left[1 \quad 0 \right] } } } , \quad { \underline { { \left[1 \quad 2 \quad 1 \right] } } } , \quad { \underline { { \left[1 \quad 2 \quad 1 \right] } } } , \quad { \underline { { \left[1 \quad - 2 \quad 0 \right] } } } . } \\ { { \underline { { 0 } } } \quad 1 \quad 2 { \mathrm { J } } } , \quad { \underline { { \left[0 \quad 0 \right] } } } , \quad { \underline { { 0 } } } { \mathrm { J } } } \end{array} }

$$A_1 = E_1^{(2)} A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & -3 \\ 4 & -2 & -4 & -10 \end{bmatrix},$$

注意：这些矩阵在仅使用初等行变换的前提下，无法进一步化简为更简单的形式。然而，若同时允许初等列变换，则可能实现进一步的化简，如下一命题所示。此处仍略去详细证明。

定理 2．任一非零的 m × n 矩阵，均可通过一系列初等行变换与初等列变换，化为如下形式之一的 m × n 矩阵：

$$A_2 = E_2^{(3)} A_1 = \begin{bmatrix} 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & -3 \\ 0 & -2 & -4 & 2 \end{bmatrix},$$

[I _ { m } 0] , \quad { \binom { I _ { n } } { 0 } } , \quad { \binom { I _ { r } } { 0 } } , \quad I _ { n }\tag{8}

$$A_3 = E_3^{(1)} A_2 = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & -2 & -4 & 2 \end{bmatrix},$$

回顾初等变换与初等矩阵之间的对应关系，并结合式（6）与式（7），我们由定理 2 可推出下述结论：

定理 3．设 A 为一个 m × n 矩阵，则存在有限个初等矩阵 E _ { 1 } , E _ { 2 } , \dots , E _ { k + s }，使得

$$A_4 = E_4^{(3)} A_3 = \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

是式 (8) 中的矩阵之一。

由于初等矩阵均为非奇异矩阵（参见习题 1），其乘积亦为非奇异矩阵；因此，式 (9) 中的矩阵可表为 P A Q 的形式，其中 P 与 Q 均为非奇异矩阵。这一观点的重要性足以使其成为一个正式命题。

推论 1. 对任意 m × n 矩阵 A，均存在一个 m × m 非奇异矩阵 P 与一个 n × n 非奇异矩阵 Q，使得 PAQ 是式 (8) 中的某一个矩阵。

现在考虑所有非零的 m × n 矩阵所构成的集合（其中 m 与 n 固定）。由于式 (8)
所列各类矩阵的总数严格有限，故必存在大量互不相同的矩阵，它们均以式 (8)
中的同一个矩阵为其相抵标准形（即“约化形式”）。
考虑其中两个这样的矩阵 A 与 B。根据定理 3，存在两组初等矩阵，分别将 A 与 B 化为同一标准形。因此

form" for A, which is widely used in solving linear systems (see Section 2.9).

Performing appropriate elementary column operations, it is not difficult to see that the matrix A can be further reduced to the *diagonal* form:

$$A_5 = A_4 E_5^{(3)} E_6^{(3)} E_7^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

where $E_5^{(3)}$, $E_6^{(3)}$, $E_7^{(3)}$ denote the following matrices of type (3):

$$E_5^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad E_6^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad E_7^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Summarizing all of these reductions, we have

$$(E_4^{(3)} E_3^{(1)} E_2^{(3)} E_1^{(2)}) A (E_5^{(3)} E_6^{(3)} E_7^{(3)}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6}$$

or

$$PAQ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{7}$$

where the matrices P and Q are products of elementary matrices and are, therefore, nonsingular. □

An application of the elimination procedure just used to the case of a generic rectangular $m \times n$ matrix leads to the following important result, which is presented without a formal proof.

Theorem 1. *Any nonzero $m \times n$ matrix can be reduced by elementary row operations to reduced row echelon form, which has the following defining properties:*

(1) *All zero rows are in the bottom position(s).*

(2) *The first nonzero element (reading from the left) in a nonzero row is a one (called a "leading one").*

(3) *For $j = 2, 3, \ldots, m$ the leading one in row j (if any) appears to the right of the leading one in row $j - 1$.*

(4) *Any column containing a leading one has all other elements equal to zero.*

又因初等矩阵的逆矩阵仍是同类型的初等矩阵（参见习题 2），故可推出

$$A_5 = A_4 E_5^{(3)} E_6^{(3)} E_7^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

因此，矩阵 A 可通过对 B 施行若干次初等变换而得到。

用推论 1 的语言来表述，我们已证明存在适当阶数的非奇异矩阵 P 与 Q，使得

$$E_5^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad E_6^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad E_7^{(3)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

下一个定理将上述结果加以总结。

定理 4. 若 A 与 B 是同型矩阵（即均为 m × n 矩阵），则下列命题彼此等价：

(1) A 与 B 具有相同的行最简形（即形式 (8)）；

(2) 其中任一矩阵均可通过对另一矩阵施行初等行变换与初等列变换而得到；

(3) 存在非奇异矩阵 P 与 Q，使得 $A = PBQ$。

证 这些命题均可作为如下“矩阵等价”概念的定义。例如，若命题 (3) 成立，则称矩阵 A 与 B 等价，记作 $A \sim B$，。

显然，关系 $A \sim B$ 是一个等价关系，即它满足：

(1) $A \sim A$（任一矩阵与其自身等价），此为自反性；

(2) 若 $A \sim B$，，则 $B \sim A$（对称性）；

(3) 若 $A \sim B$ 且 $B \sim C$，，则 $A \sim C$（传递性）。

于是，所有固定阶数 m × n 的矩阵构成的集合被划分为互不相交的、彼此等价的矩阵类（即等价类）。每个 m × n 矩阵 A 恰属于且仅属于其中一类；该类可描述为全体形如 P A Q 的矩阵之集合，其中 P 与 Q 分别取遍所有 m × m 阶与 n \times n 阶非奇异矩阵。

每一等价类均由且仅由一个形如 (8)

的矩阵唯一代表并确定；该矩阵称为该类中所有矩阵的标准形（亦称最简形）。例如，若 $r < m < n$，，则矩阵

$$(E_4^{(3)} E_3^{(1)} E_2^{(3)} E_1^{(2)}) A (E_5^{(3)} E_6^{(3)} E_7^{(3)}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

是某一等价类的代表元；它是该等价类中所有矩阵的标准型（典范形式）。显然，只要知道式 (11) 中的数 r，即可唯一确定一个等价类。这一事实为下一节对数 r，的深入研究提供了依据。

The following matrices are all in reduced row echelon form:

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Observe that these matrices cannot be reduced to simpler forms by the application of elementary row operations. If, however, elementary column operations are also permitted, then further reductions may be possible, as the next theorem shows. Again, a detailed proof is omitted.

Theorem 2. *Any nonzero $m \times n$ matrix can be reduced by the application of elementary row and column operations to an $m \times n$ matrix in one of the following forms*:

$$[I_m \quad 0], \quad \begin{bmatrix} I_n \\ 0 \end{bmatrix}, \quad \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, \quad I_n \tag{8}$$

Recalling the connection of elementary operations with elementary matrices, we derive the next theorem from Theorem 2 (see also Eqs. (6) and (7)).

Theorem 3. *Let A be an $m \times n$ matrix. There exists a finite sequence of elementary matrices $E_1, E_2, \ldots, E_{k+s}$ such that*

$$E_k E_{k-1} \cdots E_1 A E_{k+1} E_{k+2} \cdots E_{k+s} \tag{9}$$

is one of the matrices in (8).

Since elementary matrices are nonsingular (see Exercise 1), their product is also and, therefore, the matrix in Eq. (9) can be written in the form PAQ, where P and Q are nonsingular. This point of view is important enough to justify a formal statement.

Corollary 1. *For any $m \times n$ matrix A there exist a nonsingular $m \times m$ matrix P and a nonsingular $n \times n$ matrix Q such that PAQ is one of the matrices in* (8).

Consider now the set of all nonzero $m \times n$ matrices with fixed m and n. Since the number of matrices of the types in (8) is strictly limited, there must exist many different matrices having the same matrix from (8) as its reduced form.

Consider two such matrices A and B. In view of Theorem 3, there are two sequences of elementary matrices reducing A and B to the same matrix. Hence

$$E_k E_{k-1} \cdots E_1 A E_{k+1} \cdots E_{k+s} = \tilde{E}_l \tilde{E}_{l-1} \cdots \tilde{E}_1 B \tilde{E}_{l+1} \cdots \tilde{E}_{l+t},$$

本节最后给出若干补充习题。第一题提供了一种利用初等行变换求可逆矩阵之逆的方法。

习题 4. 设 A 是一个 n \times n 矩阵，则同阶矩阵

$$\begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

[A \quad I _ { n }] \qquad \mathrm { a n d } \qquad [I _ { n } \quad B]\tag{12}

$$[I_m \quad 0], \quad \begin{bmatrix} I_n \\ 0 \end{bmatrix}, \quad \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}, \quad I_n$$

当且仅当 A 可逆且 B ==A ^ { - 1 } 时，彼此行等价（即：二者可通过初等行变换相互化简得到）。

提示. 若式 (12) 中的两个矩阵行等价，则必有 \begin{array} { r l } { [A } & { { } I _ { n }] = P [I _ { n } \quad B] = } \end{array}=\begin{array} { r l } { [P } & { { } P B] } \end{array}。因此，若对 [A \quad I _ { n }] 实施高斯消元法所得结果为 [I _ { n } \quad B]，则必有 B = A ^ { - 1 }。若此类消元过程无法完成，则 A 不可逆。

习题 5. 证明：若两个方阵等价，则它们要么均非奇异（即可逆），要么均奇异（即不可逆）。

习题 6. 验证：一个 n \times n 矩阵是非奇异的，当且仅当它与单位矩阵 I _ { n } 行等价（或列等价）。因此，包含单位矩阵的互等价矩阵类，恰好由所有适当阶数的非奇异矩阵构成。

习题 7. 证明：任一非奇异矩阵均可分解为若干初等矩阵的乘积。

习题 8．证明：给定矩阵的行最简形（简化行阶梯形）是唯一的。□

2.8 矩阵的秩

设 A 是一个 m × n 矩阵。定理 2.7.2 中所述矩阵 A 的简化（标准）形中单位矩阵的阶数 r (1 ≤ r ≤ min (m , n))，称为 A 的秩，记作 r = rank A。回顾由式 (2.7.8) 给出的 A 的所有可能的标准形，我们有

$$E_k E_{k-1} \cdots E_1 A E_{k+1} E_{k+2} \cdots E_{k+s}$$

$$E_k E_{k-1} \cdots E_1 A E_{k+1} \cdots E_{k+s} = \tilde{E}_l \tilde{E}_{l-1} \cdots \tilde{E}_1 B \tilde{E}_{l+1} \cdots \tilde{E}_{l+t},$$